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Course: Calculus Beta

Assignment: MyLab Math - Uge 4

Solve the following equations for the real numbers x and y .

a. $(7 + 6i)^2 - 2(x - iy) = x + iy$

b. $\left(\frac{5 + 5i}{5 - 5i}\right)^2 + \frac{6}{x + iy} = 1 + i$

c. $(8 - 3i)(x + iy) = 7(x - 2iy) + 12i - 4$

a. To make the equation easier to solve for x and y , start by expressing each side of the equation in the form $a + ib$.

The left side of the equation includes $(7 + 6i)^2$, which can also be written as $(7 + 6i)(7 + 6i)$. To find this product, first multiply using the distributive rule.

Multiply using the distributive rule.

$$(7 + 6i)(7 + 6i) = (7)(7) + (7)(6i) + (6i)(7) + (6i)(6i)$$

Simplify each product in this expression.

$$(7)(7) + (7)(6i) + (6i)(7) + (6i)(6i) = 49 + 42i + 42i + 36i^2$$

The second and third terms in this expression are both multiples of i . Combine these like terms.

$$49 + 42i + 42i + 36i^2 = 49 + 84i + 36i^2$$

The symbol i represents a number whose square equals -1 .

In other words, $i^2 = -1$. Substitute this into the last term of the expression from earlier. Then simplify that term and combine like terms.

$$\begin{aligned} 49 + 84i + 36i^2 &= 49 + 84i + 36(-1) && \text{Substitute } i^2 = -1. \\ &= 49 + 84i - 36 && \text{Simplify the last term.} \\ &= 13 + 84i && \text{Combine like terms.} \end{aligned}$$

This means $(7 + 6i)^2 = 13 + 84i$. Substitute this into the given equation.

$$\begin{aligned} (7 + 6i)^2 - 2(x - iy) &= x + iy \\ 13 + 84i - 2(x - iy) &= x + iy && \text{Substitute } (7 + 6i)^2 = 13 + 84i. \end{aligned}$$

Rewrite the left side of this equation in the form $a + ib$.

$$\begin{aligned} 13 + 84i - 2(x - iy) &= x + iy \\ 13 + 84i - 2x + 2iy &= x + iy && \text{Apply the distributive rule.} \\ (13 - 2x) + i(84 + 2y) &= x + iy && \text{Group the real terms and the imaginary terms.} \end{aligned}$$

Each side of the equation is now in the form $a + ib$. That is, the equation now states that two complex numbers are equal. Two complex numbers are equal if and only if their real parts are equal and their imaginary parts are equal.

This means that $(13 - 2x) + i(84 + 2y) = x + iy$ if and only if $13 - 2x = x$ and $84 + 2y = y$. Since each of these two equations involves only one variable, they can be solved separately.

Solve the equation $13 - 2x = x$ for x .

$$\begin{aligned} 13 - 2x &= x \\ 13 &= 3x && \text{Add } 2x \text{ to both sides and simplify.} \\ \frac{13}{3} &= x && \text{Divide both sides by 3 and simplify.} \end{aligned}$$

Solve the equation $84 + 2y = y$ for y .

$$\begin{aligned} 84 + 2y &= y \\ 84 + y &= 0 && \text{Subtract } y \text{ from both sides and simplify.} \\ y &= -84 && \text{Subtract 84 from both sides and simplify.} \end{aligned}$$

Thus, $x = \frac{13}{3}$ and $y = -84$.

b. As in part (a), start by expressing each side of the equation in the form $a + ib$.

The left side of the equation includes $\frac{5 + 5i}{5 - 5i}$. To clear the i from the denominator, multiply the numerator and denominator by the complex conjugate of the denominator.

The complex conjugate of $z = a + ib$ is $\bar{z} = a - ib$.

This means that the complex conjugate of the denominator, $5 - 5i$, is $5 + 5i$.

Multiply the numerator and denominator of $\frac{5 + 5i}{5 - 5i}$ by $5 + 5i$ using the method of multiplying complex numbers described in part (a).

$$\begin{aligned}\frac{5 + 5i}{5 - 5i} &= \frac{(5 + 5i)(5 + 5i)}{(5 - 5i)(5 + 5i)} && \text{Multiply the numerator and denominator by } 5 + 5i. \\ &= \frac{25 + 25i + 25i - 25}{25 + 25i - 25i + 25} && \text{Use the distributive rule.} \\ &= \frac{50i}{50} && \text{Simplify the numerator and denominator.} \\ &= i && \text{Divide.}\end{aligned}$$

Substitute $\frac{5 + 5i}{5 - 5i} = i$ into the left side of the given equation. Then simplify the result and isolate the fraction on the left side.

$$\begin{aligned}\left(\frac{5 + 5i}{5 - 5i}\right)^2 + \frac{6}{x + iy} &= 1 + i \\ i^2 + \frac{6}{x + iy} &= 1 + i && \text{Substitute } \frac{5 + 5i}{5 - 5i} = i. \\ -1 + \frac{6}{x + iy} &= 1 + i && \text{Simplify } i^2. \\ \frac{6}{x + iy} &= 2 + i && \text{Add 1 to both sides.}\end{aligned}$$

It would be possible to rewrite this equation by multiplying the numerator and denominator by $x - iy$, the complex conjugate of the denominator. However, here we will instead multiply both sides of the equation by $x + iy$. Note that this is allowable because the original equation would involve division by 0 if $x + iy = 0 + i0 = 0$.

Multiply both sides of the equation by $x + iy$. Then find the product on the right side using the method of multiplying complex numbers from earlier.

$$\begin{aligned}\frac{6}{x + iy} &= 2 + i \\ 6 &= (2 + i)(x + iy) && \text{Multiply both sides by } x + iy. \\ 6 &= 2x + 2iy + ix - y && \text{Use the distributive rule.} \\ 6 + i0 &= (2x - y) + i(x + 2y) && \text{Express each side in the form } a + ib.\end{aligned}$$

Since two complex numbers are equal if and only if their real parts are equal and their imaginary parts are equal, this equation yields the following system of equations.

$$\begin{aligned}2x - y &= 6 \\ x + 2y &= 0\end{aligned}$$

Although there are many ways to solve this system, here we will use the method of elimination and eliminate y . One way to do this is to multiply the first equation by 2 and add the result to the second equation.

Multiply the first equation by 2 and simplify the result.

$$\begin{aligned}2x - y &= 6 \\ x + 2y &= 0\end{aligned} \quad \rightarrow \quad \begin{aligned}4x - 2y &= 12 \\ x + 2y &= 0\end{aligned}$$

Add these equations.

$$\begin{aligned}4x - 2y &= 12 \\ x + 2y &= 0 \\ \hline 5x &= 12\end{aligned}$$

Solving this equation for x yields $x = \frac{12}{5}$.

To find the value of y , substitute $x = \frac{12}{5}$ into one of the equations of the system from earlier. Here we will use the second equation.

$$x + 2y = 0$$

$$\frac{12}{5} + 2y = 0$$

$$2y = -\frac{12}{5}$$

$$y = -\frac{6}{5}$$

$$\text{Substitute } x = \frac{12}{5}.$$

$$\text{Subtract } \frac{12}{5} \text{ from both sides.}$$

$$\text{Divide both sides by 2.}$$

$$\text{Thus, } x = \frac{12}{5} \text{ and } y = -\frac{6}{5}.$$

c. As in parts (a) and (b), start by expressing each side of the equation in the form $a + ib$.

Express the left side of the equation in the form $a + ib$ using the method of multiplying two complex numbers from earlier.

$$(8 - 3i)(x + iy) = 7(x - 2iy) + 12i - 4$$

$$8x + 8iy - 3ix + 3y = 7(x - 2iy) + 12i - 4$$

$$(8x + 3y) + i(8y - 3x) = 7(x - 2iy) + 12i - 4$$

Use the distributive rule.

Express the left side in the form $a + ib$.

Express the right side of the equation in the form $a + ib$ by using the distributive rule to find $7(x - 2iy)$ and then grouping the real and imaginary parts of the right side.

$$(8x + 3y) + i(8y - 3x) = 7(x - 2iy) + 12i - 4$$

$$(8x + 3y) + i(8y - 3x) = 7x - 14iy + 12i - 4$$

$$(8x + 3y) + i(8y - 3x) = (7x - 4) + i(-14y + 12)$$

Use the distributive rule and simplify.

Express the right side in the form $a + ib$.

Since two complex numbers are equal if and only if their real parts are equal and their imaginary parts are equal, this equation yields the following system of equations.

$$8x + 3y = 7x - 4$$

$$8y - 3x = -14y + 12$$

Rewrite each equation so that the variables are on the left side and so that the x - and y -terms are in the same order in the two equations.

$$8x + 3y = 7x - 4$$

$$8y - 3x = -14y + 12$$

$$\rightarrow \begin{array}{rcl} x + 3y & = & -4 \\ -3x + 22y & = & 12 \end{array}$$

Although there are many ways to solve this system, here we will use the method of elimination and eliminate x . One way to do this is to multiply the first equation by 3 and add the result to the second equation.

Multiply the first equation by 3 and simplify the result.

$$x + 3y = -4$$

$$-3x + 22y = 12$$

$$\rightarrow \begin{array}{rcl} 3x + 9y & = & -12 \\ -3x + 22y & = & 12 \end{array}$$

Add these equations.

$$3x + 9y = -12$$

$$-3x + 22y = 12$$

$$\hline 31y = 0$$

Solving this equation for y yields $y = 0$.

To find the value of x , substitute $y = 0$ into one of the equations of the system from earlier. Here we will use the first equation.

$$x + 3y = -4$$

$$x + 3(0) = -4$$

$$x = -4$$

Substitute $y = 0$.

Simplify the left side.

Thus, $x = -4$ and $y = 0$.